

CURSORS

```
Database changed
mysql> CREATE TABLE Student (
  ->     SID INT PRIMARY KEY,
  ->     SName VARCHAR(50),
  ->     Marks INT
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Student VALUES
  -> (101, 'Rahul', 85),
  -> (102, 'Priya', 78),
  -> (103, 'Arun', 92),
  -> (104, 'Sneha', 88),
  -> (105, 'Kiran', 75);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+
| SID | SName | Marks |
+-----+-----+-----+
| 101 | Rahul | 85    |
| 102 | Priya | 78    |
| 103 | Arun  | 92    |
| 104 | Sneha | 88    |
| 105 | Kiran | 75    |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

SIMPLE CURSOR

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE SimpleCursor()
  -> BEGIN
  ->     DECLARE id INT;
  ->     DECLARE name VARCHAR(50);
  ->
  ->     DECLARE student_cursor CURSOR FOR
  ->         SELECT SID, SName FROM Student;
  ->
  ->     OPEN student_cursor;
  ->
  ->     FETCH student_cursor INTO id, name;
  ->
  ->     SELECT id AS StudentID, name AS StudentName;
  ->
  ->     CLOSE student_cursor;
  -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL SimpleCursor();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
| 101       | Rahul       |
+-----+-----+
1 row in set (0.01 sec)
```

CURSOR WITH LOOP

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE DisplayStudents()
```

```
mysql> BEGIN
```

```
mysql>     DECLARE id INT;
```

```
mysql>     DECLARE name VARCHAR(50);
```

```
mysql>     DECLARE done BOOLEAN DEFAULT FALSE;
```

```
mysql>
```

```
mysql>     DECLARE student_cursor CURSOR FOR
```

```
mysql>         SELECT SID, SName FROM Student;
```

```
mysql>
```

```
mysql>     DECLARE CONTINUE HANDLER FOR NOT FOUND
```

```
mysql>         SET done = TRUE;
```

```
mysql>
```

```
mysql>     OPEN student_cursor;
```

```
mysql>
```

```
mysql>     read_loop: LOOP
```

```
mysql>
```

```
mysql>         FETCH student_cursor INTO id, name;
```

```
mysql>
```

```
mysql>         IF done THEN
```

```
mysql>             LEAVE read_loop;
```

```
mysql>         END IF;
```

```
mysql>
```

```
mysql>         SELECT id AS StudentID, name AS StudentName;
```

```
mysql>
```

```
mysql>     END LOOP read_loop;
```

```
mysql>
```

```
mysql>     CLOSE student_cursor;
```

```
mysql>
```

```
mysql> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CALL DisplayStudents();
```

StudentID	StudentName
101	Rahul

```
1 row in set (0.00 sec)
```


StudentID	StudentName
102	Priya

```
1 row in set (0.01 sec)
```


StudentID	StudentName
103	Arun

```
1 row in set (0.01 sec)
```


StudentID	StudentName
104	Sneha

```
1 row in set (0.01 sec)
```


StudentID	StudentName
105	Kiran

```
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH WHILE

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CursorWhile()
-> BEGIN
->     DECLARE id INT;
->     DECLARE name VARCHAR(50);
->     DECLARE done BOOLEAN DEFAULT FALSE;
->
->     DECLARE student_cursor CURSOR FOR
->         SELECT SID, SName FROM Student;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
->         SET done = TRUE;
->
->     OPEN student_cursor;
->
->     FETCH student_cursor INTO id, name;
->
->     WHILE done = FALSE DO
->
->         SELECT id AS StudentID, name AS StudentName;
->
->         FETCH student_cursor INTO id, name;
->
->     END WHILE;
->
->     CLOSE student_cursor;
->
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL CursorWhile();
+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         101 | Rahul       |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         102 | Priya       |
+-----+-----+
1 row in set (0.00 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         103 | Arun        |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         104 | Sneha       |
+-----+-----+
1 row in set (0.01 sec)

+-----+-----+
| StudentID | StudentName |
+-----+-----+
|         105 | Kiran       |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH REPEAT

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE CursorRepeat()
```

```
mysql> BEGIN
```

```
mysql>     DECLARE id INT;
```

```
mysql>     DECLARE name VARCHAR(50);
```

```
mysql>     DECLARE done BOOLEAN DEFAULT FALSE;
```

```
mysql>
```

```
mysql>     DECLARE student_cursor CURSOR FOR
```

```
mysql>         SELECT SID, SName FROM Student;
```

```
mysql>
```

```
mysql>     DECLARE CONTINUE HANDLER FOR NOT FOUND
```

```
mysql>         SET done = TRUE;
```

```
mysql>
```

```
mysql>     OPEN student_cursor;
```

```
mysql>
```

```
mysql>     REPEAT
```

```
mysql>         FETCH student_cursor INTO id, name;
```

```
mysql>
```

```
mysql>         IF NOT done THEN
```

```
mysql>             SELECT id AS StudentID, name AS StudentName;
```

```
mysql>         END IF;
```

```
mysql>
```

```
mysql>     UNTIL done
```

```
mysql>     END REPEAT;
```

```
mysql>
```

```
mysql>     CLOSE student_cursor;
```

```
mysql>
```

```
mysql> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> CALL CursorRepeat();
```

StudentID	StudentName
101	Rahul

```
1 row in set (0.00 sec)
```

StudentID	StudentName
102	Priya

```
1 row in set (0.00 sec)
```

StudentID	StudentName
103	Arun

```
1 row in set (0.01 sec)
```

StudentID	StudentName
104	Sneha

```
1 row in set (0.01 sec)
```

StudentID	StudentName
105	Kiran

```
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.01 sec)
```

CURSOR WITH UPDATE

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE AddBonus()
  -> BEGIN
  ->     DECLARE id INT;
  ->     DECLARE done BOOLEAN DEFAULT FALSE;
  ->
  ->     DECLARE student_cursor CURSOR FOR
  ->         SELECT SID FROM Student;
  ->
  ->     DECLARE CONTINUE HANDLER FOR NOT FOUND
  ->         SET done = TRUE;
  ->
  ->     OPEN student_cursor;
  ->
  ->     read_loop: LOOP
  ->         FETCH student_cursor INTO id;
  ->
  ->         IF done THEN
  ->             LEAVE read_loop;
  ->         END IF;
  ->
  ->         UPDATE Student
  ->         SET Marks = Marks + 5
  ->         WHERE SID = id;
  ->
  ->     END LOOP;
  ->
  ->     CLOSE student_cursor;
  ->
  -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL AddBonus();
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+
| SID | SName | Marks |
+-----+-----+-----+
| 101 | Rahul | 90    |
| 102 | Priya | 83    |
| 103 | Arun  | 97    |
| 104 | Sneha | 93    |
| 105 | Kiran | 80    |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```